2024/DSE/II/8

Suppose in an economy, the public increases their cash holding by withdrawing bank deposits. Immediately after the withdrawal, the banking system of the economy faces a reserve shortage of $100 million.

The following table shows the balance sheet of the banking system immediately after the withdrawal.

Assets ($ million)		Liabilities ($ million)	
Reserves	600	Deposits	3 500
Loans	2 900		

(a) Find the required reserve ratio. (1 mark)

(b) Suppose the required reserve ratio is reduced by 5 percentage points. Calculate the maximum possible change in deposits after deposit creation/contraction. Show your workings. (3 marks)

SECTION 5: MONEY AND BANKING (II)

5.1 MONETARY BASE (M0) AND MONETARY POLICY

Multiple Choice Questions

1994/AL/II/18

When the central bank buys government bonds from the non-bank public, the immediate impact on commercial banks is that the reserves are ____________ the amount of the purchase.

A. increased by
B. increased by a portion of
C. decreased by
D. decreased by a multiple of

Study the following balance sheet of a banking system and answer questions 16 and 17.

Assets ($)		Liabilities ($)	
Reserves	4 000	Demand Deposits	8 000
Loans	3 000		
Investment	1 000		

1995/AL/II/16

Suppose the banking system has $2 000 excess reserves. When the central bank buys $2 000 worth of government bonds from the non-bank public, the excess reserves of the banking system would increase by

A. $0.
B. $1 000.
C. $1 500.
D. $2 000.

1995/AL/II/17

If the banks engage in credit creation after the purchase of government bonds by the central bank, which of the following statements will be true?

(1) The maximum amount of demand deposits will be $24 000.
(2) The loans will increase.
(3) The excess reserves will decrease.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. All of the above

1995/AL/II/28

Which of the following events will **NOT** increase the reserves of a commercial bank?

A. A customer deposits a cheque with the bank.
B. The central bank purchase a government bond from one of the bank's depositors.
C. The bank purchases a government bond from another commercial bank.
D. The bank borrows from the central bank.

1996/AL/II/19
The total amount of currency issued is $100 billion. Half is held by the non-bank public and half is held by commercial banks as reserves. Suppose the minimum reserve ratio is 20% and banks do not keep excess reserves, the money supply will be

A. $60 billion.
B. $100 billion.
C. $250 billion.
D. $300 billion.

For Question 15, consider the following balance sheet of a banking system.

Assets ($)		Liabilities ($)	
Reserves	3 000	Deposits	12 000
Loans	4 000		
Investment	5 000		

Suppose the banking system has $600 excess reserves, the public does not hold any cash and deposits all cash into the banking system.

1999/AL/II/15
Suppose the central bank sells $500 worth of government bonds to the commercial banks, the money supply in the economy will

A. decrease by $500.
B. decrease by $2 500.
C. remain unchanged.
D. increase by $500.

2001/AL/II/13
When the central bank purchases government bonds from the commercial banks,

A. the reserve of the banking system will decrease.
B. the required reserves of the banking system will decrease.
C. the banking multiplier will increase.
D. the maximum loan that the banking system can create will increase.

2002/AL/II/21
Study the following balance sheet of a banking system and answer question 21. Suppose the required reserve ratio is 25%.

Assets ($)		Liabilities ($)	
Reserves	3 000	Deposits	8 000
Loans	5 000		

Newly printed bank notes worth $400 are issued resulting in a $200 increase in cash held by the non-bank public and a $1 000 increase in the money supply. The total amount of excess reserves in the banking system will be

A. $800
B. $950
C. $1 000
D. $1 600

Study the following balance sheet of a banking system and answer Questions 19 and 20.

Assets ($)		Liabilities ($)	
Reserves	2 000	Deposits	10 000
Loans	7 000		
Investment	1 000		

2005/AL/II/19
Suppose the banking system does not hold any excess reserve. The immediate effect(s) of a withdrawal of $500 cash from the banking system is/are

(1) the reserve shortage is $100.
(2) the money supply remains unchanged.
(3) the monetary base remains unchanged.

A. (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2005/AL/II/20
After the credit contraction process is completed, the money supply will decrease by ________ and the monetary base will ________.

A. $1 500 ...... decrease by $500
B. $1 500 ...... remain unchanged
C. $2 000 ...... decrease by $500
D. $2 000 ...... remain unchanged

2007/AL/II/7
The monetary base will increase if

A. the central bank issues more banknotes.
B. the commercial banks hold more reserves.
C. the central bank reduces the required reserve ratio.
D. All of the above.

2011/AL/II/20
Which of the following will lead to a fall in monetary base?

A. a fall in the velocity of circulation of money.
B. an increase in the required reserve ratio
C. an open market sale of government bonds by the central bank.
D. All of the above

2014/DSE/I/33
Study the following information about an economy.

Monetary base	$500 billion
The coins and notes circulated in the public	$300 billion
Required reserve ratio	20%

If all the banks are fully loaned up, the amount of deposits will be __________.

A. $1 000 billion
B. $1 500 billion
C. $2 500 billion
D. $4 000 billion

2015/DSE/I/29
Study the following balance sheet of a banking system. The required reserve ratio is 20%.

Assets ($ million)		Liabilities ($ million)	
Reserves	300	Deposits	1 000
Loans	700		

If the central bank issues newly printed banknotes worth $200 million, the cash held by the non-bank public will increase by $100 million and the money supply will increase by $500 million. The total amount of excess reserves in the banking system will then be ____________.

A. $100 million
B. $120 million
C. $280 million
D. $400 million

2015/DSE/I/32
The monetary base increases when

A. the commercial banks hold less excess reserves.
B. the commercial banks buy government bonds from the public.
C. the central bank buys government bonds from the public.
D. the central bank increases the discount rate.

2018/DSE/I/32
Refer to the following information about an economy.

Money supply	$35 000
Actual reserve ratio	25%
Excess reserves	$1150

Suppose the public always holds $12 000 cash. If all banks loan out their excess reserves and all the loans are re-deposited into the banking system,

A. the maximum possible amount of money supply will be $28 750.
B. the monetary base of the economy will be $16 600.
C. the maximum possible amount of loans will be $23 000.
D. the maximum banking multiplier will be 4.

1990/AL/II/03
The most feasible and frequently used instrument of monetary policy in Hong Kong is:

A. open market operations in foreign exchange
B. change in discount rates
C. variations in reserve requirements
D. moral suasion

1990/AL/II/04
Which of the following are pre-conditions for open market operations to be effective?

(1) There is a large stock of government securities.
(2) Commercial banks have a propensity to keep excess reserves.
(3) There is a well-developed securities market.
(4) There is a Central Bank.

A. (1), (2) and (3) only.
B. (1), (2) and (4) only.
C. (1), (3) and (4) only.
D. (2), (3) and (4) only.

1991/AL/II/23
Other things being equal, an open market sale of bonds by the central bank to the non-bank public will

(1) curb the credit creation power of the commercial banks.
(2) raise the market rate of interest.
(3) increase the amount of securities held by the commercial banks.
(4) lead to a fall in national income.

A. (1), (2) and (3) only
B. (1), (2) and (4) only
C. (1), (3) and (4) only
D. (2), (3) and (4) only

1992/AL/II/04
If the demand for money increases and the monetary authorities want the interest rate to remain unchanged, which of the following policies should be recommended?

A. raising the discount rate.
B. buying bonds in the open market.
C. raising the legal minimum reserve ratio.
D. None of the above.

1993/AL/II/16
Which of the following has a contractionary effect on the money supply?

A. There is an increase in the cash drain ratio.
B. There is a decrease in the rediscount rate.
C. There is a decrease in the legal reserve ratio.
D. The government buys bonds in the open market.

1996/AL/II/18
Which of the following are examples of contractionary policy?

(1) an increase in the tax rate
(2) a reduction in the discount rate
(3) a sale of government bonds by the central bank

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. All of the above.

1997/AL/II/08
Assuming banks are fully loaned up, the money supply will decrease when the government ________ bonds and ________ the reserve ratio.

A. sells ...... increases
B. buys ...... increases
C. sells ...... reduces
D. buys ...... reduces

1998/AL/II/17
Suppose the demand for money increases, other things begin equal, which of the following measures can stabilize the interest rate?

(1) a fall in the discount rate
(2) a fall in the required reserve ratio
(3) an open market sale of government bonds

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2000/AL/II/08
Suppose banks are fully loaned up. Which of the following will lead to a decrease in money supply?

(1) an increase in reserve ratio
(2) a reduction in the public's desire to hold cash
(3) an open market sale of government bonds by the central bank to the non-bank public

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2003/AL/II/12
A central bank can increase the money supply through

(1) an open market purchase of government bonds
(2) a reduction in the discount rate
(3) a drop in the required reserve ratio

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. All of the above.

2004/AL/II/04
Which of the following are examples of an expansionary monetary policy?

(1) a reduction in the required reserve ratio
(2) an increase in discount rate
(3) an open market purchase of government bonds by the central bank

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2005/AL/II/17
A rise in the discount rate will

A. reduce the money supply because it increases the cash held by the public.
B. reduce the money supply because it reduces the reserves borrowed by commercial banks from the central bank.
C. increase the money supply because it raises the ability of commercial banks to create loans.
D. increase the money supply because it reduces the reserves held by commercial banks.

2006/AL/II/07
Which of the following may cause an increase in the money supply?

(1) Banks loan out their excess reserves.
(2) There is a decrease in the required reserve ratio.
(3) A commercial bank purchases government bonds from another commercial bank.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2007/AL/II/10
Which of the following are examples of a contractionary monetary policy?

(1) an open market sale of bonds by the central bank
(2) an increase in the discount rate
(3) a devaluation of domestic currency

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2010/AL/II/18
When the central bank sells government bonds to the public, ______________ will increase.

A. the general price level
B. the interest rate
C. money supply
D. money demand

2012/AL/II/14
When the central bank keeps purchasing bonds from the market, which of the following is **LEAST** likely to occur?

A. The market interest rate will rise.
B. The real output level will rise.
C. The general price level will rise.
D. The asset price will rise.

2012/DSE/I/30
On-line shopping has become more popular, and hence people hold less cash for transaction purposes. Which of the following measures can be adopted by a central bank to reduce the effect of the above situation on the interest rate?

(1) an increase in discount rate
(2) an increase in required reserve ratio
(3) an open market purchase of government bonds

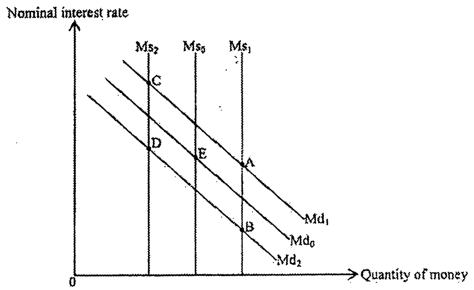
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2013/DSE/I/28
An expansionary monetary policy is adopted in China when

(1) the Ministry of Finance (財政部) provides cash subsidies to agricultural workers.
(2) the People's Bank of China (中國人民銀行) reduces the required reserve ratio of all commercial banks.
(3) the Bank of China lowers its lending rate to small and medium-sized enterprises.

A. (1) only
B. (2) only
C. (3) only
D. (1), (2) and (3)

2017/DSE/I/32
The diagram below shows the money demand and supply curves of an economy. The original equilibrium point is at E.



Suppose the risk of holding interest bearing assets (such as bonds) increases and the central bank reduces the discount rate. The new equilibrium point will be at _______.

A. A
B. B
C. C
D. D

2017/DSE/I/37
Which of the following policies would most likely lead to a drop in aggregate output?

A. The government bought bonds from the public and raised the tax allowance at the same time.
B. The government sold bonds to the public and reduced transfer payment to the poor at the same time.
C. The government reduced required reserve ratio and raised the progressivity of income tax at the same time.
D. The government increased the discount rate and lowered profits tax rate at the same time.

2019/DSE/I/41
Study the following table.

Policy I	Policy II	Policy III
• an open market purchase of government bonds • a rise in the discount rate	• an open market purchase of government bonds • a rise in the salaries tax allowance	• a rise in the salaries tax allowance • a rise in the discount rate

Which of the policies in the above table is/are expansionary?

A. Policy I only
B. Policy II only
C. Policy III only
D. Policy I and Policy III only

2020/DSE/I/30
Refer to the following information about an economy. Suppose all banks do not hold any excess reserves.

	$ million
Currency in public circulation	10
Monetary base	30
Money supply	110

If the public deposits all the cash into the banking system,

A. the actual banking multiplier will be 4.
B. the maximum amount of loans will be $120 million.
C. the maximum amount of deposits will be $140 million.
D. the new money supply will be $160 million.

2020/DSE/I/32
Which of the following is a contractionary monetary policy?

A. an increase in the discount rate
B. an increase in the salaries tax rate
C. a decrease in the required reserve ratio
D. a decrease in the quota for migration

2021/DSE/I/34

Suppose the actual inflation rate and the nominal interest rate are −1% and 3% respectively. Which of the following statements are correct?

(1) The cost of holding cash is 3%.
(2) The expected real interest rate is higher than the nominal interest rate.
(3) The actual real rate of return on holding cash is 1%.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2021/DSE/I/35

Which of the following would offset the effect on aggregate output of an expansionary fiscal policy?

A. The public held more cash.
B. The use of electronic wallets became more popular.
C. The interbank rate decreased.
D. The government increased quota on the importation of foreign domestic helpers.

2023/DSE/I/32

Which of the following will lead to an increase in monetary base?

A. The central bank raises the discount rate.
B. The central bank reduces the required reserve ratio.
C. The central bank buys government bonds from the public.
D. The central bank replaces some of the paper banknotes in circulation with polymer banknotes.

2024/DSE/I/28

Suppose the central bank of Country A lowers the discount rate. Which of the following situations would weaken the effect of the above policy on the aggregate output of Country A in the short ran?

A. The use of digital money becomes more popular in Country A.
B. Some major trading partners of Country A experience an economic recession.
C. The government of Country A relaxes the restrictions on the importation of foreign workers.
D. The government of Country A increases its investment on local infrastructure.

2024/DSE/I/35

The table below shows the monetary data of a country. The public always holds a fixed amount of cash.

Money supply	$95 million
Cash held by public	$15 million
Actual reserve ratio	25%
Excess reserves	$4 million

Based on the above information, which of the following statements are correct?

(1) The monetary base is $35 million.
(2) The maximum banking multiplier is 5.
(3) When all banks are fully loaned up, the maximum amount of money supply is $115 million.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2024/DSE/I/39

A contractionary monetary policy is adopted in China when

(1) the People's Bank of China *(中國人民銀行)* increases the required reserve ratio of commercial banks.
(2) the People's Bank of China cuts the interest rate on its loans to commercial banks,
(3) the Chinese Government raised the tax rates of the individual income tax.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

Short & Structured Questions

1994/AL/II/3

(a) Explain briefly how purchases of government bonds by the central bank would affect the money supply. (3 marks)

(b) If the reserve ratio is 0.2, what is the impact on the money supply if the central bank purchases bonds worth 1 million dollars? List **TWO** of the assumptions you have made in your calculation. (5 marks)

1996/AL/II/5(a)

Explain how the sale of government bonds by the central bank to the public affects the money supply. (4 marks)

2004/AL/II/1(c)

Name one similarity and one difference between each pair of the following terms.

monetary base (or high-powered money); money supply. (3 marks)

2007/AL/II/3(a)

What is meant by the monetary base? What is meant by the money supply? Why is money supply often a multiple of base money? When will the banking multiplier equal unity? (5 marks)

2008/AL/II/1(c)

Name one similarity and one difference between the following pairs of terms:

discount rate; interbank rate. (3 marks)

2012/DSE/II/13

The following table shows the balance sheet of the banking system of an economy.

Assets ($ million)		Liabilities ($ million)	
Reserves	500	Deposits	2 000
Loans	1 500		

Suppose the public does not hold cash and there are $100 million excess reserves in the banking system.

(a) Find the required reserve ratio. Show your working. (2 marks)

(b) Find the maximum possible amount of deposits if the banks are fully loaned up. Show your working. (2 marks)

(c) The central bank injects $200 million newly-printed banknotes into the banking system. If the banks lend out all of their excess reserves, what will the amount of money supply be in the economy? Show your working. (4 marks)

2013/DSE/II/12

The following table shows the balance sheet of the banking system of an economy.

Assets ($ million)		Liabilities ($ million)	
Reserves	400	Deposits	2 000
Loans	1 600		

The public holds $1 000 million cash and all banks are fully loaned up.

(a) Calculate the monetary base and the money supply in the economy. Show your workings. (4 marks)

(b) If the required reserve ratio is adjusted to 25%, calculate the new money supply in the economy. Show your workings. (2 marks)

2014/DSE/II/7(b)

In an attempt to reduce its growing fiscal deficits, the US government has tried to increase tax on the one hand and decrease fiscal spending on the other. But many worry that such a policy would give rise to undesirable side effects.

An alternative policy is to finance the fiscal deficits by printing money. Briefly discuss **ONE** advantage and **ONE** disadvantage of this alternative policy on the US economy as opposed to the above fiscal policy. (4 marks)

2014/DSE/II/12

The Prime Minister of Japan, Shinza Abe, adopted a series of policy measures in 2013 to expand the Japanese economy. One of the policy measures was stepping up the scale of the purchase of government bonds and other financial assets by the central bank.

(a) If the Japanese central bank purchases government bonds from Japanese commercial banks, explain whether the monetary base of Japan would be affected. (3 marks)

(b) Describe the process of deposit creation/contraction which may result from the purchase of government bonds by the central bank. Explain whether the process would further affect the monetary base of Japan. (5 marks)

2015/DSE/II/6

Study the following balance sheet of the banking system in Country A.

Assets ($ million)		Liabilities ($ million)	
Reserves	1 000	Deposits	5 000
Loans	4 000		

Suppose all banks are fully loaned up and never hold excess reserves. The public always holds $150 million in cash.

(a) Calculate the following items of this banking system:

(i) legal reserve ratio (1 mark)

(ii) money supply (2 marks)

A firm in Country A has received $50 million from abroad and deposited the sum of money into a bank.

(b) Calculate the new monetary base in Country A. Show your workings. (2 marks)

(c) Calculate the new money supply in Country A. Show your workings. (3 marks)

2016/DSE/II/8

The following is the balance sheet of a banking system.

Assets ($ million)		Liabilities ($ million)	
Reserves	1 000	Deposits	4 000
Loans	3 000		

Suppose the public always holds $500 million of cash and the banking system does not hold excess reserves.

(a) Calculate the monetary base and money supply. (2 marks)

The central bank increases the required reserve ratio to 50%.

(b) Calculate the change in monetary base and the change in money supply. Show your working. (4 marks)

(c) (i) Why is money supply often a few times more than monetary base? (3 marks)

(ii) Under what situation will money supply be equal to monetary base? (1 mark)

2018/DSE/II/7(a)
The following is the balance sheet of a banking system.

Assets ($ million)		Liabilities ($ million)	
Reserves	500	Deposits	2 000
Loans	1 500		

Initially the banks held $100 million excess reserves and the public held $100 million cash. Suppose the central bank buys $40 million worth of government bonds from the public.

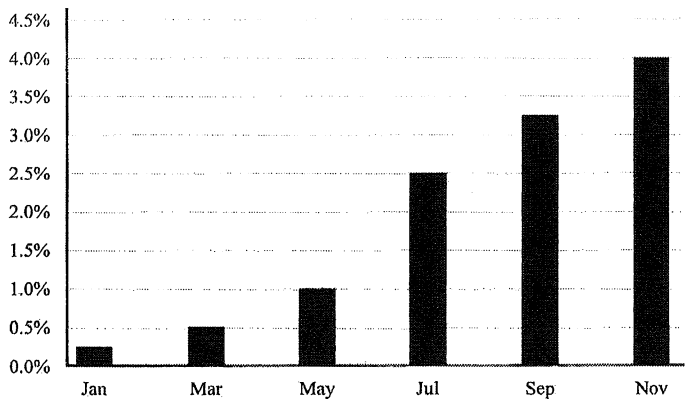
(i) Calculate the new monetary base. Show your workings. (2 marks)

(ii) State **TWO** assumptions under which the actual change in money supply would equal the maximum possible change in money supply. (2 marks)

(iii) Calculate the maximum possible change in money supply. Show your workings. (4 marks)

2023/DSE/II/12c

Source D: Discount rate of the Federal Reserve* of the US from January to November 2022



*Federal Reserve is the central bank of the US.

(c) Refer to Source D. How may the change in discount rate as shown in the source help relieve the pressure on rising price level? Without using a diagram, explain your answer in the light of aggregate supply and/or aggregate demand. (5 marks)

2025/DSE/II/10(Cross Topic 5.1 /5.2 /5.3 /6.1)

The monetary policy of the US receives much attention. (a) The table below shows some data on the US economy.

	May 2022	May 2023
Money supply (USD Billion)	21691.3	20 837.4
Inflation rate (measured by year-on-year percentage change in implicit price deflator of GDP)	7.0%	3.6%

(i) Calculate the money supply growth rate in the US from May 2022 to May 2023. (1 mark)

(ii) Based on the above data, what was the real output growth rate in the US from May 2022 to May 2023 as implied by the quantity theory of money? Explain your answer. (3 marks)

The Federal Reserve (the central bank of the US) began to lower the discount rate in September 2024.

(b) **WITHOUT** using a diagram, explain how the equilibrium nominal interest rate would be affected by the lowering of the discount rate. (3 marks)

(c) With the aid of an AS-AD diagram, explain how the real output in the US would be affected in the short run by the lowering of the discount rate. (5 marks)

(d) To achieve the same effect on the real output as in part (c), an economist suggested that the US Government should increase its expenditure on infrastructure instead of having the Federal Reserve lower the discount rate. Discuss the suggestion by giving **ONE** argument **FOR** and **ONE** argument **AGAINST** it. (4 marks)